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Evaluation of the efficacy of empagliflozin in patients with preserved left ventricular ejection fraction undergoing transcatheter aortic valve implantation for aortic stenosis: protocol for a randomised, open-label, controlled study

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ABSTRACT

Introduction Heart failure occasionally develops after transcatheter aortic valve implantation (TAVI) for severe aortic stenosis (AS), despite procedural success. Most cases present with mildly reduced or preserved left ventricular ejection fraction (LVEF), underscoring the role of diastolic dysfunction. Sodium-glucose cotransporter 2 (SGLT2) inhibitors have shown benefits across the heart failure spectrum, independent of LVEF. The purpose of this randomised controlled trial is to determine whether adding a SGLT2 inhibitor to conventional medications improves LV diastolic function in patients with preserved LVEF after TAVI.

Methods and analysis This study is a prospective, single-centre, open-label, randomised, parallel-group, two-arm trial enrolling patients with mildly reduced or preserved LVEF (≥40%) undergoing TAVI for severe AS. Participants will be randomised in a 1:1 ratio to receive either conventional medications plus empagliflozin or conventional medications alone. In the empagliflozin group, participants will receive conventional medical therapy plus empagliflozin 10 mg orally once daily, initiated 4 weeks after TAVI. Empagliflozin treatment will continue throughout the study period. Participants in the control group will receive conventional medications without empagliflozin. The primary endpoint is the change in E/e', assessed by echocardiography from treatment initiation at 4 weeks post TAVI (day 1) to day 168 (week 24). Each group will include 50 patients, totalling 100 patients.

Ethics and dissemination Ethical approval for this study has been obtained from the Chiba University Hospital Certified Clinical Research Review Board (CRB0111-25).

Trial registration number jRCT1031250190.

STRENGTHS AND LIMITATIONS OF THIS STUDY

⇒ This randomised, open-label, parallel-group trial will investigate the efficacy of sodium-glucose cotransporter 2 inhibitors in patients undergoing transcatheter aortic valve implantation (TAVI) with severe aortic stenosis and mildly reduced or preserved left ventricular ejection fraction.
⇒ The primary endpoint was the change in E/e' assessed by echocardiography from 4 weeks to 28 weeks after TAVI.
⇒ Secondary endpoints include additional echocardiographic parameters, functional status, biomarkers and safety outcomes to support interpretation of the primary endpoint.
⇒ Limitations of this trial include that the primary endpoint is only one marker of diastolic function and clinical or functional outcomes are not assessed.

INTRODUCTION

Transcatheter aortic valve implantation (TAVI) is increasingly used for patients with severe aortic stenosis (AS), and its non-inferior or better clinical outcomes compared with surgical aortic valve replacement have been reported in many clinical trials.[1–3] Nevertheless, heart failure following TAVI remains common, with a reported 1-year incidence of approximately 12–24%.[4–6] Therefore, there remains a significant proportion of patients with poor prognosis even after successful TAVI. On the other hand, reduced left ventricular ejection fraction (LVEF) is

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observed in only about one-third of patients undergoing TAVI.[4 7 8] As the majority of patients undergoing TAVI have mildly reduced or preserved LVEF, left ventricular (LV) diastolic dysfunction is likely to be involved in the development of heart failure after TAVI. Thus, assessment of LV diastolic function and appropriate treatment for these patients are crucial.

Based on the results of previous large-scale randomised controlled trials evaluating the use of sodium-glucose cotransporter 2 (SGLT2) inhibitors, these agents have emerged as an important therapeutic option for patients with heart failure, regardless of ejection fraction.[9–12] These landmark studies, however, excluded individuals with severe valvular heart disease requiring intervention, leaving a gap in the evidence base for patients who experience recurrent heart-failure events after recovery from conditions such as AS.

Recently, a randomised trial showed that adding an SGLT2 inhibitor to standard care in high-risk patients undergoing TAVI significantly reduced the composite of all-cause death and worsening heart failure compared with standard care alone.[13] However, because that study included patients with reduced LVEF, the findings do not specifically reflect outcomes in patients with mildly reduced or preserved LVEF, who represent the majority of heart failure cases after TAVI. In addition, no trial to date has specifically examined whether SGLT2 inhibitors improve LV diastolic function in patients with mildly reduced or preserved LVEF after TAVI. Therefore, evidence for SGLT2 inhibitors in the post-TAVI population remains limited. The present trial is designed to address this evidence gap under clinical equipoise with appropriate safety monitoring.

METHODS

Study design

In this phase IV, prospective, single-centre, open-label, randomised, parallel-group, two-arm study, participants will be randomised 1:1 to receive either conventional medications plus empagliflozin, an SGLT2 inhibitor or conventional medications only. This trial uses a superiority framework.

Eligibility criteria

Eligible patients are those who meet all inclusion criteria mentioned below and none of the listed exclusion criteria. Inclusion criteria are: (1) patients with AS and symptoms of heart failure requiring TAVI based on the guidelines for management of valvular heart disease[14 15]; (2) patients scheduled for TAVI at Chiba University Hospital; (3) patients aged between 70 years and 90 years at the time of enrolment[16]; (4) patients with a LVEF ≥40%; (5) patients who have provided written informed consent to participate in this study (if the patient cannot provide a signature, a family member/designate may sign on their behalf).

Exclusion criteria include any of the following: (1) acute heart failure (New York Heart Association (NYHA) class IV); (2) current use of SGLT2 inhibitors; (3) allergy, hypersensitivity or previous intolerance to SGLT2 inhibitors; (4) chronic cystitis and/or recurrent urinary tract infections occurring more than twice a year; (5) a patient with difficulty in oral intake of food; (6) severe renal dysfunction (estimated glomerular filtration rate (eGFR) <20 mL/min/1.73 m² or dialysis at baseline); (7) a patient without written informed consent; (8) a patient whom the investigator considers to be ineligible as a subject.

Recruitment

Recruitment of this study started in August 2025 and will end in January 2029, or until a total of 100 participants have been recruited. This study is being conducted at Chiba University Hospital.

Patient and public involvement

Patients and/or the public were not involved in the design, conduct, reporting or dissemination plans of this research.

Sample size calculation

The target sample size for this randomised trial is 100. The aim of this study is to investigate the effects and mechanisms of empagliflozin and to compare the empagliflozin group (combination of conventional drugs and empagliflozin) with the control group (conventional drugs only) in improving LV diastolic function in patients with AS undergoing TAVI, with E/E′ set as the primary endpoint.

A previous investigation in patients with type 2 diabetes mellitus reported that over 168 days (6 months) the mean (±SD) change in E/e′ (ΔE/e′) was 0.7±4.1 in the control arm and −1.7±3.4 in the SGLT2 inhibitor arm.[17] Assuming a common SD of 4.0 and an expected between-group difference of 2.4 in ΔE/e′, a significance level of 5% (α=0.05) and a two-sided t-test with 80% power (1−β = 0.8), 45 patients per group is required (90 in total). Considering approximately 10% of discontinuation or dropout, the total sample size of 100 patients is required for this study.

Registration and assignment methods

In principle, (1) registration should be completed within 7 days of the procedure. (2) The principal investigator or subinvestigators will obtain written consent and confirm that participants meet the selection criteria and do not violate the exclusion criteria as a result of the screening test. Participants deemed eligible by the principal investigator or subinvestigators will be enrolled prior to the start of study drug administration. Participants will be enrolled in the study by faxing or bringing the case registration form (CRF) with them. (3) The principal investigator and subinvestigators will complete the CRF and bring it to the data management centre at Chiba Clinical Research Center. The data centre will enter the necessary information for case registration on the data system and confirm the results of eligibility determination and

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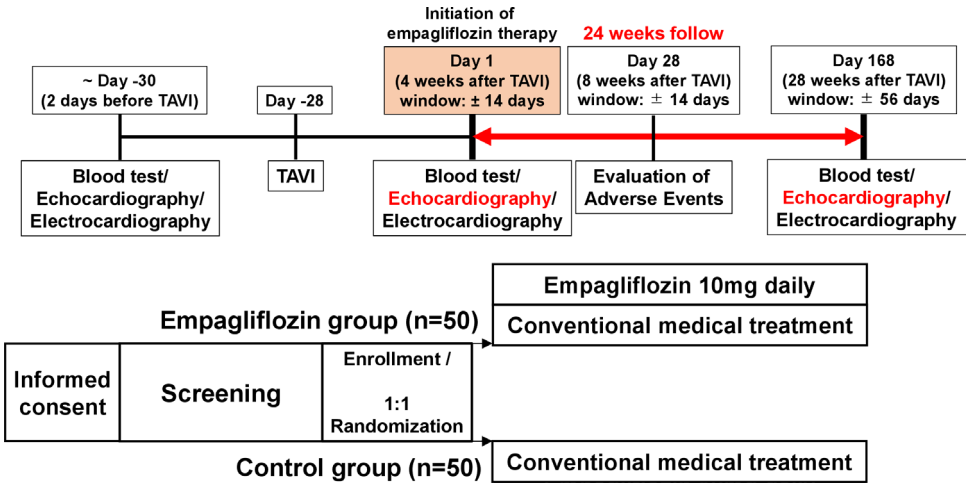


Figure 1 Graphic outline of study design. Patients are randomly assigned to either the empagliflozin group with conventional medications plus empagliflozin or the control group with conventional medications only. TAVI, transcatheter aortic valve implantation.

assignment on the system. If the patient is found to be eligible, the investigator or subinvestigators will start the protocol treatment according to the results of the assignment. (4) Once a patient is registered, the registration will not be cancelled (deleted from the database). In case of duplicate registration, the initial registration information (registration number) will be used in all cases. In case of registration errors or duplicate registrations, the data centre shall be notified immediately.

Assignment adjustment factors

Participants will be randomly assigned 1:1 to the study to receive conventional medications plus empagliflozin or conventional medications only. Eligible patients will be randomised to either the empagliflozin group or the control group at a ratio of 1:1, by employing a minimisation method balancing for sex (male or female), age (<85 years or ≥85 years) and LVEF (<50% or ≥50%).

Study procedures

The study outline is shown in figure 1. A patient who meets the inclusion criteria and provides written consent before TAVI is eligible. Randomisation is performed within 7 days after the procedure to assign them to the empagliflozin or control group. In the empagliflozin group, participants will receive conventional medical therapy plus empagliflozin 10 mg orally once daily, initiated 4 weeks after TAVI (defined as day 1; visit window: ±14 days). In the control group, participants will receive conventional medications without empagliflozin. Adverse events will be evaluated on day 28 (visit window: ±14 days), and a comprehensive follow-up visit on day 168 (visit window: ±56 days) will include a physical examination, monitoring vital signs, laboratory tests, transthoracic echocardiography, electrocardiography and a review of concomitant medications. Empagliflozin will be continued at least until day 168. Adherence will be checked on day 28 and day 168. No post-trial provision of empagliflozin is planned. All echocardiographic measurements are performed under blinded conditions by cardiologists certified as Board Certified Fellows of the Japan Society of Ultrasonics in Medicine, according to the American Society of Echocardiography (ASE) and the European Association of Cardiovascular Imaging (EACVI) guidelines.[18 19]

Discontinuation or modification criteria

Treatment will be interrupted if continuation is considered inappropriate by the principal investigator or subinvestigators, or if severe adverse events occur (dehydration, acute kidney injury, hypoglycaemia, acidosis or urinary tract infection). Treatment may be restarted once the condition improves and resumption is judged feasible.

Participants will be withdrawn from protocol treatment and the study if any of the following apply: at the day 1 assessment (4 weeks post TAVI) an exclusion criterion is identified; adverse events make initiation or continuation difficult and discontinuation is judged necessary; consent is withdrawn; continued participation is deemed impossible; the principal investigator or the trial steering committee terminates the study.

Concomitant SGLT2 inhibitors are prohibited. If heart failure worsens, rescue therapy will use diuretics other than SGLT2 inhibitors and the study will continue; use of an SGLT2 inhibitor as rescue therapy in the control group will lead to discontinuation of protocol treatment. Participants will remain in the study, and follow-up will be continued under the intention-to-treat (ITT) principle.

Blinds

Blinds are not performed.

Endpoint

The primary endpoint of the study is the change in E/e′ (ΔE/e′) assessed by echocardiography from treatment initiation at 4 weeks post TAVI (day 1) to day 168 (week 24). The secondary efficacy endpoints include: (1) change in echocardiographic parameters at day 168 (week 24) after treatment initiation—change in left atrial volume

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index (LAVi), change in left ventricular mass index, change in systolic pulmonary artery pressure, change in tricuspid annular plane systolic excursion, change in tricuspid lateral annular systolic velocity and change in right ventricular fractional area change; (2) N-terminal pro–B-type natriuretic peptide (NT-proBNP) value and change after treatment initiation (ΔNT-proBNP); (3) NYHA functional class at day 168; (4) change in body weight at day 168; (5) cardiovascular events (death or hospitalisation for heart failure). The secondary safety endpoints include the incidence of adverse events—dehydration, acute kidney injury, hypoglycaemia, acidosis and urinary tract infection. As an exploratory endpoint, we evaluate the prevalence of E/e′ in three categories at day 168 (week 24): E/e′ <8, E/e′ 8–14 and E/e′ >14.

Statistical analysis

The analyses of the primary and secondary endpoints will be performed according to the intention-to-treat principle in the full analysis set (FAS), which includes all patients who: received at least one dose of treatment after randomisation, have at least one post-baseline efficacy assessment and meet all key eligibility criteria. Patients who have major eligibility violations will be excluded from the FAS. The per protocol set will consist of patients from the FAS, excluding those with significant protocol deviations such as unmet inclusion or exclusion criteria, use of prohibited concomitant medications (any SGLT2 inhibitors other than empagliflozin in the empagliflozin group; any SGLT2 inhibitors except as rescue therapy in the control group) or receipt of concomitant therapies inconsistent with the protocol. For the baseline characteristics, summary statistics will be presented according to the type of variable. Continuous variables will be described using appropriate measures such as the mean, SD, median, IQR and ranges. Categorical variables will be summarised by frequency and percentage for each group. The primary endpoint—ΔE/e′ will be compared between the two groups (SGLT2 inhibitor, control) using a two-sample t-test. Secondary endpoints will be analysed similarly to provide additional insights into the primary analyses. Multiplicity adjustments are not planned at this stage. For safety endpoints, adverse events will be summarised by frequency and percentage of patients. If necessary, two-sided 95% CIs (Clopper-Pearson method) and Fisher's exact test will be provided. Statistical significance will be set at a two-sided α=0.05, and 95% CIs will be calculated. This study is designed without interim efficacy analyses, and there are currently no plans to impute missing data. All analyses will be performed using SAS software, V.9.4 (SAS Institute, Cary, North Carolina, USA).

Quality control

The monitoring manager is responsible for ensuring that the human rights of the subjects are protected, the study is conducted in accordance with the protocol and the data are accurately collected. Central monitoring will be conducted on a regular basis based on data from case report forms obtained at the data centre. The monitoring manager shall prepare a report that includes a summary of important findings such as diseases, nonconformities or other facts and shall report the results of such monitoring to the principal investigator. The independent data monitoring committee, which consists of at least three expert members independent of the study, will conduct safety monitoring, including comparison of the incidence of adverse events between the study treatment and control group, and detailed examination of serious adverse events, as necessary, for the purpose of ensuring patient safety. The principal investigator will make a decision to discontinue the study as a whole if it is considered difficult to continue the study due to unforeseen adverse events, illness or other factors.

Ethics committee

This study will be conducted in accordance with the Declaration of Helsinki (revised 2024). Ethical approval for this study has been obtained from the Chiba University Hospital Certified Clinical Research Review Board (CRB0111-25) and the study is registered in the Japan Clinical Trials Registry (jRCT1031250190). The investigators will explain the concept of this study to patients and obtain written informed consent from all participating patients. The initial version of the protocol was approved on 16 June 2025, and patient recruitment began on 7 August 2025.

Informed consent

All participants will receive adequate information about the nature, purpose, possible risks and benefits of the trial, and on alternative therapeutic choices using informed consent approved by the institutional review board (IRB) (a blank copy of the participant information sheet and consent form used in this study is provided as online supplemental material 1). A participant must be given ample time and opportunity to ask questions and to consider participation in the trial. A completed informed consent is required for enrolment in the trial. The investigators must maintain the original signed consent form and a copy of the signed consent form. To assure patient confidentiality, trial participants will be allocated a unique trial identification number for reference throughout the trial.

Dissemination

The findings of this trial will be disseminated through peer-reviewed publications and conference presentations and will also be disseminated to participants.

DISCUSSION

TAVI has been shown to provide effective haemodynamic relief in patients with AS. However, most patients undergoing TAVI are elderly and face persistent residual risks, including atrial fibrillation, chronic kidney disease, potentially unrecognised cardiac amyloidosis and other

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comorbidities.[4] Additionally, procedural complications such as conduction disturbances, paravalvular leaks and prosthesis-patient mismatch may lead to heart failure even in cases with preserved LVEF after TAVI.[4] Therefore, appropriate post-TAVI management is crucial to prevent heart failure when clinically indicated.

SGLT2 inhibitors improve outcomes irrespective of LVEF and are now recommended for early initiation.[9–12] In post-TAVI patients, these agents may provide additional benefit by exerting cardioprotective effects through integrated haemodynamic, metabolic, anti-inflammatory and cardiovascular protective mechanisms.[20–22] Despite these biologically plausible benefits, major heart failure trials excluded patients with significant valvular disease,[9–12] and concerns about genitourinary infections or ketoacidosis have often limited the use of SGLT2 inhibitors in frail TAVI populations.

The DAPA-TAVI trial, one of the randomised trials conducted in this setting, showed that dapagliflozin reduced the composite of death or heart failure worsening at 12 months.[13] However, key structural and diastolic parameters, such as the E/e′, have not been reported. Moreover, because the DAPA-TAVI trial enrolled patients with reduced LVEF, its findings offer limited insight into the dominant post-TAVI heart-failure phenotype, heart failure with mildly reduced or preserved ejection fraction. To address these gaps, we have designed a prospective, randomised controlled trial restricted to post-TAVI patients with LVEF ≥40%. The primary endpoint is the change in E/e′ from baseline to 6 months; secondary endpoints include the change in LAVi and other echocardiogram markers. By focusing on diastolic function, our study will test whether SGLT2 inhibition can mechanistically mitigate mildly reduced or preserved LVEF physiology after TAVI.

In this study, E/e′ is set as the primary endpoint. E/e′ provides a reliable non-invasive estimate of LV filling pressures and has been validated against invasive measurements.[23] Although cardiac MRI may offer more comprehensive structural and tissue characterisation, routine cardiac MRI assessment in all participants is not feasible in daily clinical practice because of limited availability, cost and patient-related constraints. Several randomised and prospective studies have also used E/e′ as a primary endpoint.[17 24 25] Therefore, E/e′ was selected to allow comparison with prior literature and to ensure external validity. According to ASE/EACVI recommendations, an average E/e′<8 suggests normal filling pressures, whereas an average E/e′>14 has high specificity for elevated filling pressures.[19] Based on these clinically relevant cut-offs, we evaluate the prevalence of E/e′ in three categories at day 168 (week 24): E/e′<8, E/e′ 8–14 and E/e′>14 as an exploratory endpoint.[26]

Safety is a central consideration when prescribing empagliflozin. A meta-analysis of randomised controlled trials demonstrated no significant increase in urinary tract infections with SGLT2 inhibitors versus the control group.[27] Similarly, the DAPA-TAVI trial found no differences between groups in urinary tract infections, bacteremia or syncope.[13] These findings support the safety of empagliflozin in the post-TAVI population. However, SGLT2 inhibitors are associated with a higher incidence of genital mycotic infections.[13 27] Accordingly, our trial follow-up and management protocols ensure the early detection and treatment of such events in all participants.

If this study demonstrates the effectiveness of empagliflozin in preventing diastolic dysfunction, the next step may be to investigate its ability to reduce heart failure events in a larger-scale study focused on patients with mildly reduced or preserved LVEF.

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Contributors SO and HK designed the original concept. All coauthors contributed significantly to the conception and design of the study, with specific additional contributions from each coauthor within their area of expertise. The protocol was written by SO and HK, and it was critically reviewed by MS, TS, KMiy, HYag, HYam, TK, MK, HS, NSE, HT, KMat, YO, HH, GM and YK. All authors gave approval for the publication. YK is the guarantor.

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